A Giant Planet Vanished -- and Won't Be Back for Decades

From: Popular Science Mined: 2026-11-12 3 min read

For the first time, **astronomers** watched a giant planet vanish. The '**hot Jupiter**' called TOI-2535 b sits about 1,000 **light-years** from Earth, but it wobbled out of view instead of crossing its star.

Astronomers first spotted it in April 2019 using NASA's TESS telescope, which finds planets by watching for tiny dips in a star's brightness. TOI-2535 b is a **gas giant** like Jupiter, but it circles its star in 8.25 days Mercury takes 88 days.

Over seven years, researchers noticed its crossings arrived as much as six hours earlier or later than predicted. Then, in March, the planet simply did not show up at all.

Researchers believe a neighboring giant planet with a tilted orbit is tugging on it, tipping it out of line with Earth. Samuel Yee of the University of California, Los Angeles, called it the most extreme example researchers have seen.

The moon's tilted orbit works the same way, which is why we do not see an **eclipse** every month. TOI-2535 b will not cross its star again for another 30 to 50 years.

Word Treasure

astronomers -- Astronomers are scientists who study the stars, planets, and everything in space.

hot Jupiter -- A hot Jupiter is a giant planet like Jupiter that sits very close to its star, so it is very hot.

gas giant -- A gas giant is a huge planet made mostly of gas, like Jupiter.

light-years -- A light-year is the distance light travels in one year, used to measure how far away stars are.

eclipse -- An eclipse happens when one thing in space passes in front of another and blocks its light.

Background you need

A telescope in space can find planets around other stars by watching for a tiny dip in a star's light when a planet crosses in front of it.

Planets circle their stars in a steady path, so scientists can predict when each crossing should happen.

If a nearby planet pulls on it, the crossing can shift or stop lining up with Earth, so the planet seems to vanish.

Break it down

WHO: Astronomers, including Samuel Yee of the University of California, Los Angeles
WHAT: The giant planet TOI-2535 b wobbled out of view and stopped crossing its star
WHERE: Around a star about 1,000 light-years from Earth
WHY: A neighboring giant planet with a tilted orbit is tugging on it and tipping it out of line with Earth

Why it matters

Watching a planet vanish teaches you that space is always moving and changing, and that careful watching can reveal things no one has ever seen before.

Test what you remember

1. What is TOI-2535 b?

(A) A hot Jupiter, a gas giant like Jupiter
(B) A star like the sun
(C) A small rocky moon
(D) A comet

2. How did astronomers first spot TOI-2535 b?

(A) Using NASA's TESS telescope
(B) Using a weather balloon
(C) Using a microscope
(D) Using a submarine

3. How long does TOI-2535 b take to circle its star?

(A) 30 days
(B) 8.25 days
(C) 88 days
(D) 365 days

4. What happened to the planet in March?

(A) The planet exploded
(B) The planet doubled in size
(C) The planet simply did not show up at all
(D) The planet crashed into its star

5. Why do researchers think the planet vanished?

(A) A cloud covered the telescope
(B) The planet turned into a star
(C) A neighboring giant planet with a tilted orbit is tugging on it
(D) Its star went out

6. When will TOI-2535 b cross its star again?

(A) Next week
(B) In one year
(C) In 30 to 50 years
(D) Never again

Answer key (try the quiz first!): 1: A 2: A 3: B 4: C 5: C 6: C

Questions to think about

1. Samuel Yee of the University of California, Los Angeles: This is the most extreme example researchers have seen.

2. Researchers: A neighboring giant planet with a tilted orbit is tugging on TOI-2535 b and tipping it out of line with Earth.

MY THOUGHTS (at least 20 words):